

ENGN1560 Transmission Lines Lab #1

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Introduction

The goal of this lab was to design a 50Ω microstrip transmission line on an FR4 board, simulate it in Ansys HFSS, and then build and measure a physical version on a vector network analyzer (VNA). The board is 70 mm wide, and 100 mm long, with a substrate thickness of 1.6 mm and a relative permittivity of 4.4. The trace is made of 0.1 mm thick copper tape, and the design frequency is 1 GHz.

The trace width was first calculated from the standard microstrip synthesis equations and checked against an online calculator. The resulting geometry was then modeled in HFSS from 0.5 to 1.5 GHz. Finally, a physical board was built, and its S-parameters and VSWR were measured, and these measurements were compared with the simulation.

Hand Calculations

To find the trace width needed to achieve a 50Ω characteristic impedance, standard closed-form microstrip equations were evaluated using the board dielectric constant of $\epsilon_r = 4.4$ and substrate thickness of $d = 1.6$ mm. First, the immediate variable A was computed using the following equation:

$$A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right)$$

Plugging in $Z_0 = 50\Omega$, and $\epsilon_r = 4.4$ gives $A = 1.52986$. Next, the width to height ratio W/d was calculated using the equation $W/d < 2$:

$$\frac{W}{d} = \frac{8e^A}{e^{2A} - 2}$$

Substituting $A = 1.52986$ yields $W/d = 1.91186$. Multiplying this ratio by the substrate thickness $d = 1.6$ mm results in a calculated trace width of $W = 3.06$ mm.

This is confirmed through the Transmission Line Online calculator with results depicted below:

Er: 4.4	Tanδ: 0 GHz	Rho: 1
Height: 63 mils	Thickness: 0 mils	Frequency: 1 GHz

Analysis/Synthesis Values

Width: 120.49395 mils	Zo: 50 Ohms	
Length: 1613.02744 mils	Angle: 90 degrees	

Analyze

Synthesize

εeff:	Loss: 0.005 dB	
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Figure 1: *Transmission Line Calculator results synthesizing a 50 Ω line on 1.6 mm FR4 ($\epsilon_r = 4.4$) at 1 GHz. The width is shown to be 120.49395 mils, which converts to 3.06 mm, confirming our hand calculations. The agreement confirms that 3.06 mm is the correct width to use in the simulation and on the physical board.*

This width matters because it sets the line’s characteristic impedance. A microstrip’s impedance depends on its inductance and capacitance per unit length, and both are controlled by how wide the trace is compared with its height above the ground plane. A wider trace sits closer to the ground plane over a larger area, which adds capacitance and lowers the impedance. A narrower trace does the opposite and raises it. The goal is 50 Ω because the VNA, cables, and SMA connectors in the lab are all designed for 50 Ω. When a signal reaches a point where the impedance changes, part of it reflects back towards the source instead of continuing to the load. A line matched to 50 Ω avoids these reflections, so nearly all the power sent into it reaches the other end.

Transmission Line Simulation

The microstrip was modeled in Ansys HFSS following the handout procedure. The ground plane was drawn as a 70 x 100 mm rectangle at $z = 0$ and assigned a perfect E boundary. The substrate was drawn as a 70 x 100 x 1.6 mm box and assigned the FR4_epoxy material. The trace was drawn as a copper box 3.06 mm wide, 100 mm long, and 0.1 mm thick, starting at $x = 33.5$ mm so that it sits centered at the top of the substrate. A 50Ω modal lumped port was placed at the end of the trace, spanning from the top of the copper down to the ground plane. The whole board was then enclosed in an air box assigned a Radiation boundary. The solution was set up at 1 GHz with a maximum of six adaptive passes and a convergence target of $\Delta S = 0.02$. A fast frequency sweep was then run from 0.5 to 1.5 GHz. The completed model is shown in Figure 2.

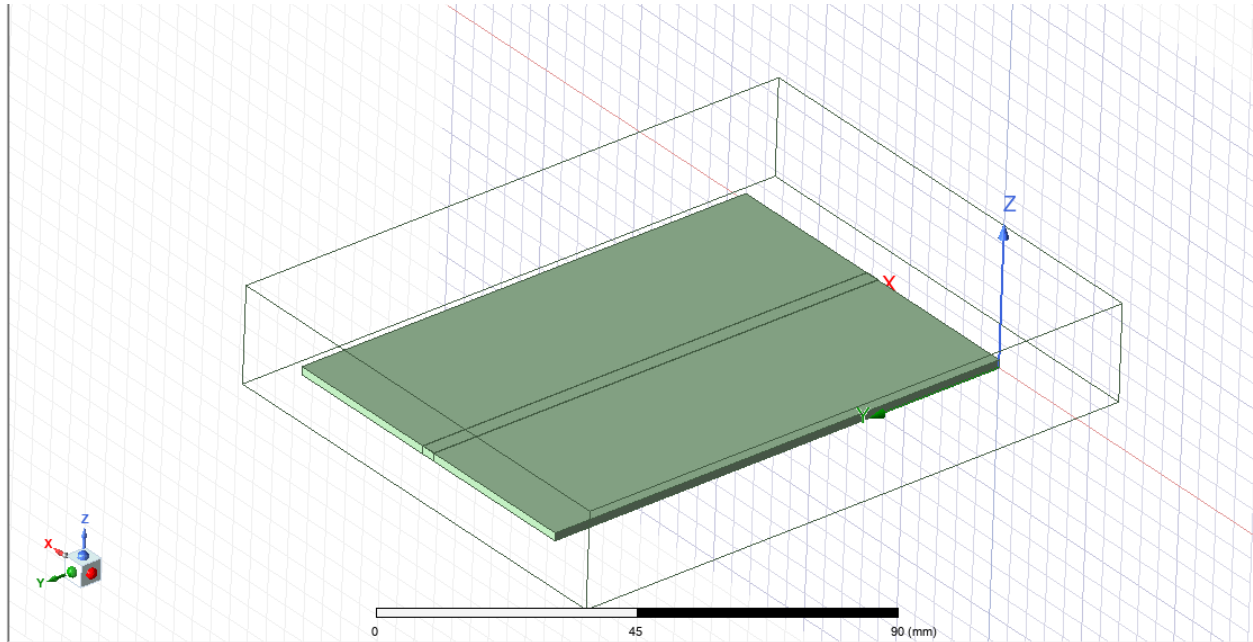


Figure 2: Completed HFSS model of the microstrip line, showing the copper trace on the FR4 substrate with lumped ports at each end, enclosed in a radiation box.

The simulated S-parameters are shown in Figure 3. S21 describes how much of a signal entering port 1 arrives at port 2, and S11 describes how much is reflected back into port 1. On the decibel scale used in the plot, 0 dB corresponds to all of the power, -10 dB corresponds to 10%, and -20 dB corresponds to 1% (reflection %).

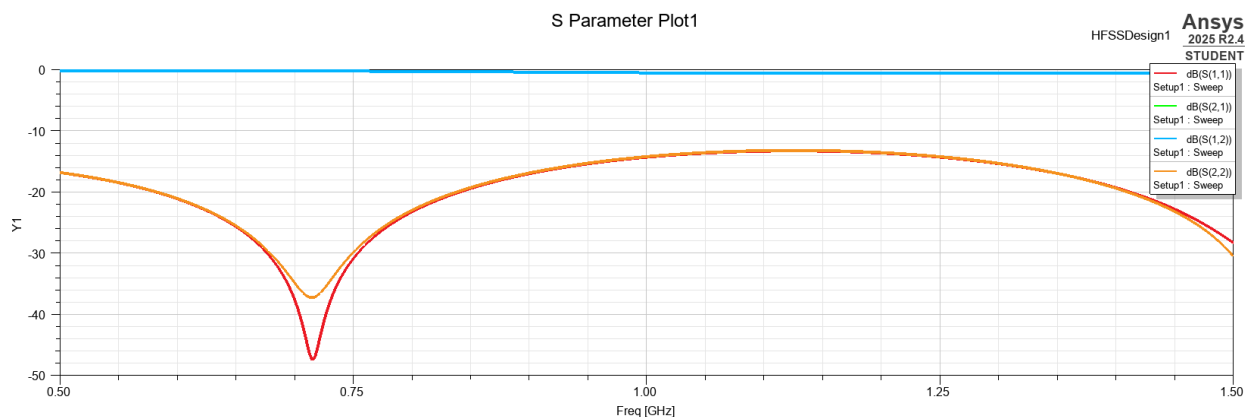


Figure 3: Simulated S-parameters in dB from 0.5 to 1.5 GHz. S21 and S12 overlap near 0 dB, while S11 and S22 form the lower curves seen above.

The transmission curves S21 and S12 stay essentially at 0 dB across the entire band, which means that nearly all of the power entering the line reaches the other end. S11 is about -17 dB at 0.5 GHz and falls to about -47 dB near 0.715 GHz. It then rises to about -13.5 dB near 1.13 GHz before dropping back to about -28 dB at 1.5 GHz. At the 1 GHz design frequency, S11 is approximately -14 dB.

These results show that the line is well matched. A line is generally well matched when S_{11} is below -10 dB, meaning less than 10% of the power reflects. The simulated line stays below that level across the entire sweep. Even at its worst frequency, only about 4.5% of the power reflects, so about 95% continues down the line. S_{21} stays near 0 dB across the band, so nearly all of that power reaches the other port. The line behaves almost like an ideal lossless 50Ω connection, which confirms that the calculated width of 3.06 mm gives an impedance close to 50Ω . The same holds when the signal is sent from the other side of the line, since S_{22} and S_{12} closely follow S_{11} and S_{21} .

The shape of the S_{11} curve, which dips and rises as frequency changes, comes from small reflections at the two ends of the line. If the line's impedance is not exactly 50Ω , a small part of the signal reflects at each port. These two reflections travel different distances before they combine, so depending on the frequency they either cancel each other out or add together. The deep dip near 0.715 GHz is where the two reflections cancel almost completely, and the peak near 1.13 GHz is where they add together. Even at that peak, the reflection stays well below -10 dB, so the mismatch causing it is small.

In Person Testing

The physical line was built on a 70×100 mm FR4 board by cutting a strip of copper tape about 3 mm wide and applying it down the center of the board. SMA connectors were attached at both ends, with the center pin of each connector soldered to the trace and the connector body connected to the ground plane on the back of the board. Before measuring, the VNA was calibrated over 950 MHz to 1.05 GHz. Calibration removes the effects of the cables themselves, so the measurement reflects only the board and its connectors. The board was then connected between the two VNA ports and measurements were taken.

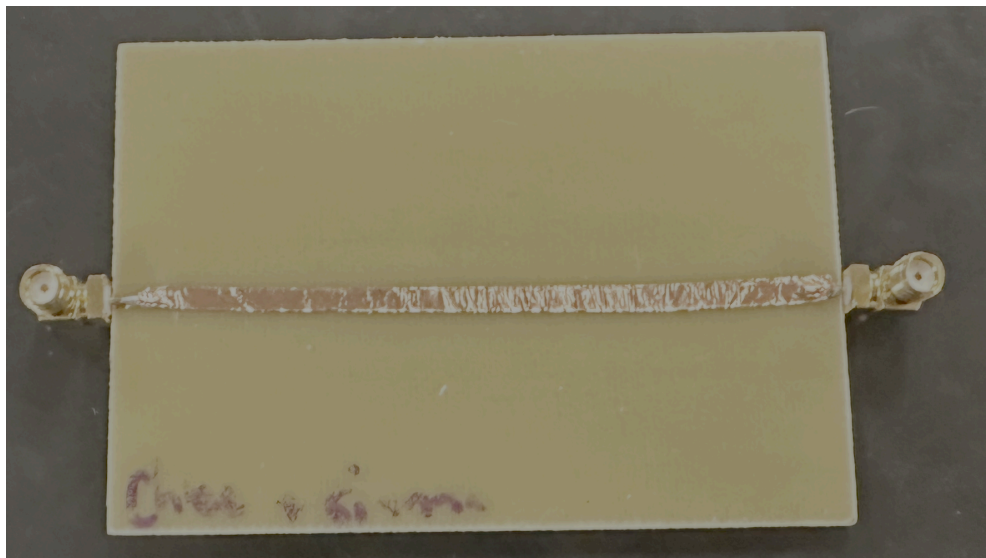


Figure 4: *Fabricated microstrip line on FR4, with a copper tape trace running the length of the board, and SMA connectors at each end.*

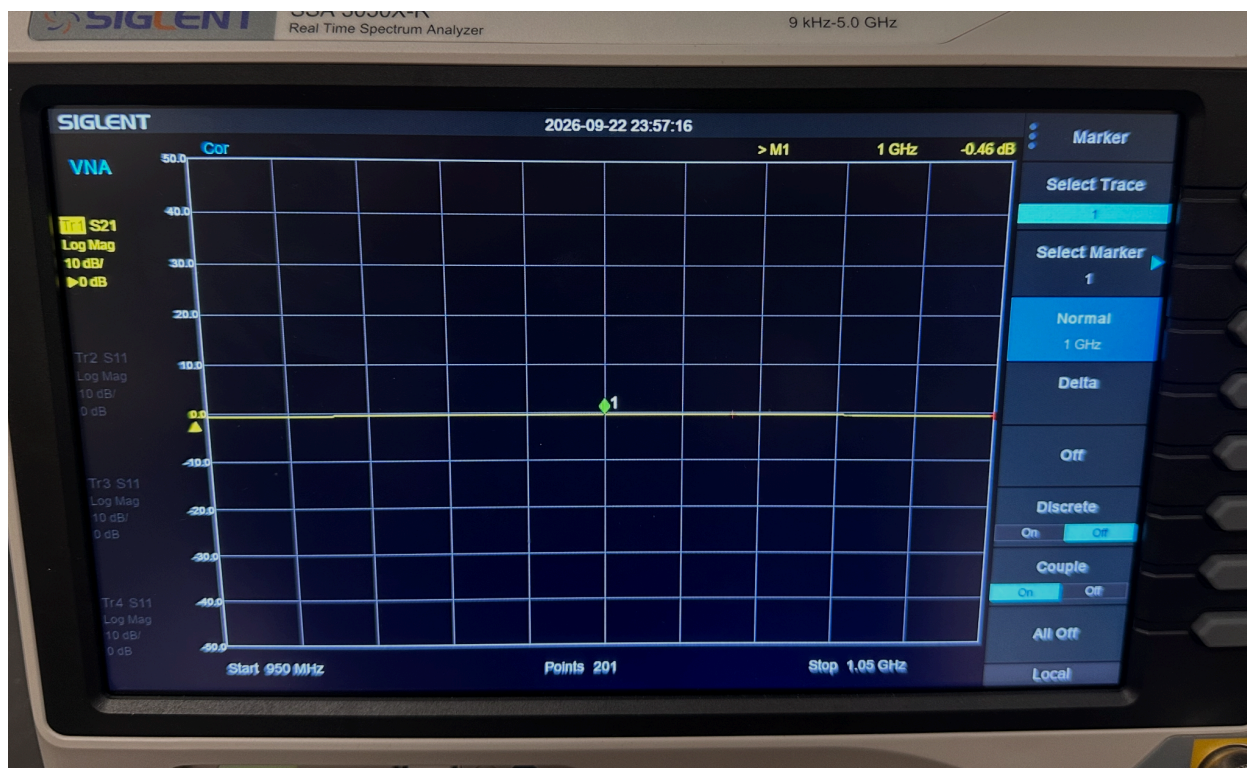


Figure 5: Measured S_{21} from 950 MHz to 1.05 GHz. Marker 1 reads -0.46 dB at 1 GHz.

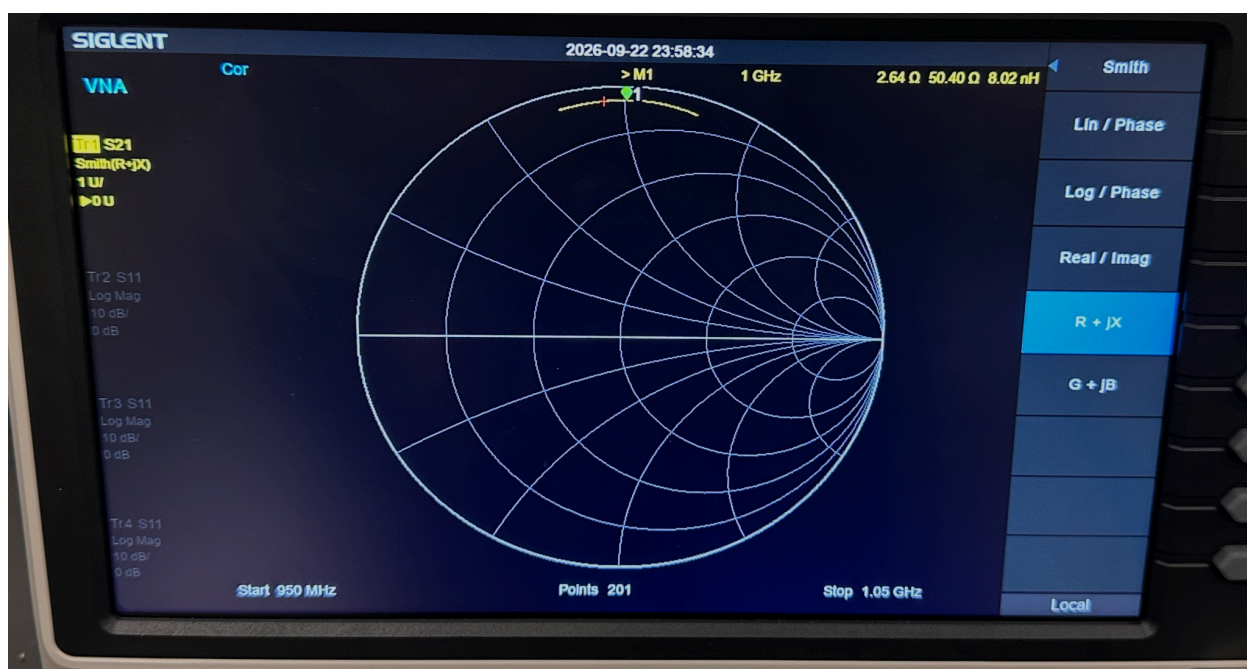


Figure 6: The same S_{21} measurement displayed on a Smith chart, showing magnitude and phase at 1 GHz.



Figure 7: *Measured VSWR at 1 GHz, taken in a follow-up session without calibration. Marker 1 reads 1.56, corresponding to roughly 5% of the incident power reflected.*

The measured S21 at 1 GHz was -0.46 dB, which means about 90% of the power sent into port 1 reached port 2. The trace stays flat at this level across the entire 100 MHz sweep, so the line performs consistently near its design frequency. On the Smith chart in Figure 6, the marker sits close to the outer edge of the chart. The outer edge represents a magnitude of 1, so a point near it confirms that almost all of the signal passes through the line.

The remaining 10% of the power was either reflected back toward the source or lost within the board. To separate these, the VSWR was measured in a follow-up session, giving a value of 1.56 at 1 GHz. This corresponds to an S11 of about -13 dB, meaning roughly 5% of the power reflected. A VSWR below 2 indicates a good match, and this agrees closely with the simulated S11 of about -14 dB at 1 GHz, which corresponds to a VSWR of about 1.5. No calibration kit was available during the follow-up session, so this reading is uncalibrated and includes some effect from the test cables. Together, the measurements suggest that about 90% of the power passed through the line, about 5% was reflected, and the remaining 5% or so was lost within the board as heat or radiation.

Comparison of Results

The measured line performed close to the simulation. Reflection was nearly the same: about 5% of the power reflected on the bench at 1 GHz, compared with about 4% in the simulation. The main difference was loss. In the simulation, essentially none of the power was lost, so about 96% reached port 2. On the bench, only about 90% arrived, meaning roughly 5% of the power was absorbed or radiated by the board. This extra loss is most likely explained by differences between the ideal model and the real board.

The largest contributor is probably loss in the FR4 itself. The simulation assumed a perfectly lossless substrate, but real FR4 absorbs a small amount of energy as heat at microwave frequencies. Over a 100 mm line at 1 GHz, this is enough to account for a few tenths of a dB.

The SMA connectors are a second difference. The simulation fed the line through ideal ports attached directly to the trace, while the real board relies on connectors and solder joints. Each transition adds a small reflection and some loss that the simulation never modeled.

The copper tape trace is also visibly imperfect, as Figure 4 shows. The tape is wrinkled along its length and bows slightly rather than running perfectly straight, so its width varies along the board. Since the width sets the impedance, these variations create small mismatches along the line, which add reflection. The wrinkled surface also makes the tape a slightly worse conductor than the smooth copper in the simulation.

Finally, the two measurements only overlap between 950 MHz and 1.05 GHz. The simulation predicted a flat S21 in this range, and the measurement confirms this. Behavior outside this band, such as the dips and peaks in the simulated S11, was not tested.

Conclusion

A 50 Ω microstrip line on 1.6 mm FR4 requires a trace width of 3.06 mm, a result confirmed by the hand calculation and an online calculator. The HFSS simulation showed that this line is well matched and nearly lossless from 0.5 to 1.5 GHz, with no more than about 4.5% of the power reflected anywhere in the band. The fabricated board measured an S21 of -0.46 dB at 1 GHz, meaning about 90% of the power was transmitted, and the response was flat across 950 MHz to 1.05 GHz. The measured VSWR of 1.56 at 1 GHz closely matched the simulated value of about 1.5, confirming that the physical line was well matched. The additional loss compared with the simulation is most likely due to FR4 dielectric loss, the SMA connector transitions, and the uneven hand-applied copper tape.